

Review

Clinical, Research, and Educational Applications of ChatGPT in Dentistry: A Narrative Review

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Abstract: Artificial intelligence (AI), specifically Generative Pre-trained Transformer (GPT) technology, has revolutionized various fields, including medicine and dentistry. The AI model ChatGPT, developed by OpenAI, mimics human language on a large scale, generating coherent and contextually appropriate responses. ChatGPT serves as an auxiliary resource for diagnosis and decision-making across various medical disciplines. This comprehensive narrative review aims to explore how ChatGPT can assist the dental sector, highlighting its potential to enhance various aspects of the discipline. This review includes a literature search on the application of ChatGPT in dentistry, with a focus on the differences between the free version, ChatGPT 3.5, and the more advanced subscription-based version, ChatGPT 4. Specifically, ChatGPT has proven to be effective in enhancing user interaction, providing fast and accurate information and improving the accessibility of knowledge. However, despite these advantages, several limitations are identified, including concerns regarding the accuracy of responses in complex scenarios, ethical considerations surrounding its use, and the need for improved training to handle highly specialized queries. In conclusion, while ChatGPT offers numerous benefits in terms of efficiency and scalability, further research and development are needed to address these limitations, particularly in areas requiring greater precision, ethical oversight, and specialized expertise.

Keywords: artificial intelligence; generative pre-trained transformer (GPT); machine learning; ChatGPT; computer-assisted diagnosis; educational technology; natural language processing; dentistry; decision making



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1. Introduction

Artificial intelligence (AI) is a field of computer science that involves developing algorithms capable of performing tasks that would normally require human intelligence [1]. These tasks include recognizing patterns and images, making decisions based on complex data, understanding and translating languages, and the ability to respond to questions or voice commands [2,3].

One of the most significant developments in AI application in recent years has been the creation and release of ChatGPT by OpenAI <https://openai.com> (accessed on 1 October 2024) (San Francisco, CA, USA). ChatGPT is a large language model based on Generative Pre-trained Transformer (GPT) technology, which uses deep learning techniques to produce text that can mimic human style on a large scale. ChatGPT was launched as a trial version in November 2022 and later made publicly available, gaining immediate attention for its ability to generate coherent and contextually appropriate responses across various fields of knowledge and everyday life [4]. The technology behind ChatGPT, and more

generally all GPT models, operates through what is known as transfer learning. Initially, the model is pre-trained on a vast dataset of publicly available texts; this phase is not specific to a task but aims to develop a general understanding of language. Subsequently, it undergoes a refinement process, during which the model is further trained on a more specific text corpus (for example, medical texts) to make its responses more appropriate to certain applications [5,6]. Furthermore, the AI algorithm uses machine learning (ML) and natural language processing techniques to identify meaningful patterns and understand user needs [7]. On the other end, the user applies prompt engineering, a process that guides AI systems to produce desired results. Specifically, it involves writing, refining, and optimizing the starting data (prompts) to encourage AI systems to create specific outputs.

Generative AI systems are designed to generate specific outputs based on the quality of the instructions provided, thus following the rule that quality instructions determine quality results, minimizing bias. The objective is to create queries that help generative AI models understand the language and every nuance to reach a specific goal. Of course, creating unbiased prompts is crucial for ensuring the fairness and effectiveness of the tool. This can be achieved by keeping the prompt clear and specific, providing sufficient context and balancing specificity and generality. Knowing if the training dataset is diverse enough would also reduce the risk of biases.

The applications of AI are numerous, from autonomous vehicle driving to customizing online shopping experiences, advanced robotics, and financial analysis [8]. AI has shown remarkable potential by being applied across all branches of the medical sector: in 2019, a team of researchers from MIT (<https://www.mit.edu> accessed on 1 October 2024) and Harvard Medical School (<https://hms.harvard.edu> accessed on 1 October 2024) developed an AI algorithm that could predict the risk of developing breast cancer up to five years before it clinically manifests, using mammograms and analyzing subtle patterns in breast tissue that are invisible to the human eye [9]. Different artificial intelligence models were developed using ultrasound data, patient histories, and medical records to diagnose polycystic ovary syndrome, allowing for earlier and more accurate diagnosis compared to traditional methods based solely on clinical and hormonal analysis [10]. Systems have been developed that integrate AI to assist doctors in choosing the most effective cancer treatments based on a vast database of the medical literature, clinical guidelines, and patient records; these systems have been used in various clinical settings to personalize oncology therapies [11]. ChatGPT can serve as an auxiliary resource for diagnosis and decision-making across various disciplines, such as cardiology, radiology, and urology, or the psychiatric field [12,13]. These examples demonstrate how AI is already a fundamental component in modern medical practice, improving diagnostic accuracy, optimizing treatments, and saving lives through quicker identification of critical conditions. The integration of these advanced technologies in the healthcare sector represents one of the most promising frontiers for the medicine of the future [7].

Similarly, like all fields of medicine, dentistry is also undergoing a digital revolution driven by continuous technological progress and the use of AI among various new tools. Oral and maxillofacial radiology benefits from AI integration in automatic tooth and bone segmentation, identification of dental implant systems, artifact reduction, and pathology detection [14–18]. It has also been shown to be helpful in improving digital impression precision, in assisting in clinical photographic diagnosis, and, generally speaking, in the fields of 3D digital dentistry (i.e., CAD/CAM, 3D printing, 3D scanning) [19–24].

Regarding GPTs, while real-world applications are under scrutiny, there are several initiatives aimed at evaluating their precision, practicality, and potential uses, taking into account their versatility. Therefore, this comprehensive narrative review aims to explore how ChatGPT can assist the entire dental sector, analyzing the potential of this technology and the ongoing studies on its use for clinical, research, and educational purposes.

2. Materials and Methods

In this review, a systematic approach was adopted to identify relevant studies. The literature search was conducted using the following electronic databases: Ovid MEDLINE, PubMed, and Web of Science. No searches of other databases or gray literature were performed. The search was conducted up to November 2024. The following search string was used: “ChatGPT” AND “Dentistry”.

2.1. Eligibility Criteria

The full texts of all possibly relevant research papers were chosen, considering as inclusion criteria only peer-reviewed articles that specifically discussed the applications of ChatGPT in all possible dental applications.

Clinical trials, observational studies, reviews, case reports, and case series were considered adequate study designs to ensure the inclusion of a broad range of evidence.

The exclusion criteria that were considered were as follows:

- Opinion pieces, editorial commentaries, and letters to editor.
- Studies focused on a specific medical specialty.
- Papers not focused on AI language models.
- Studies not in the English language.
- Papers without available full text.

2.2. Data Extraction and Review Process

Two reviewers (F.P. and G.L.G.) performed data extraction using a standardized extraction form. Findings were compared and arranged thematically. The extracted data included study characteristics, objectives, key findings, and conclusions. Any discrepancies between the two reviewers were resolved through discussion. In cases where agreement could not be reached, a third senior reviewer (R.L.G.) was consulted to ensure an unbiased resolution. A flow chart of paper selection and screening is shown in Figure 1.

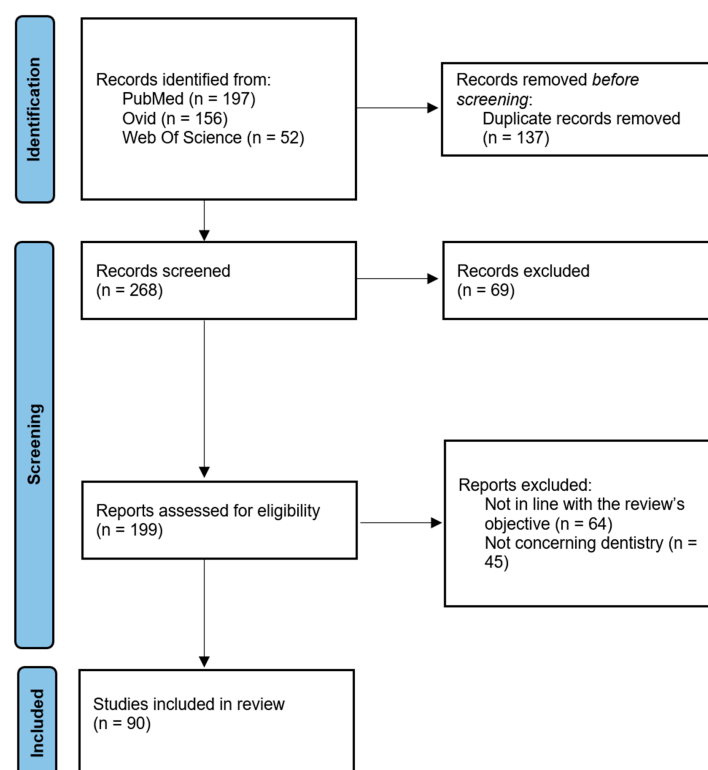


Figure 1. Papers selection and screening flow chart.

3. Results

The search engines yielded a total of 405 results. Duplicate studies were excluded, resulting in 268 remaining studies. Among these, 69 articles were excluded because they did not fit into the study design inclusion criteria. Analyzing the abstracts of 199 studies, 64 were discarded because they did not align with this review's objective or they did not pertain to ChatGPT or any other AI program, and 45 were excluded because they related to a specific field of medicine (cardiology, orthopedic surgery, neurology, etc.). Finally, 90 articles underwent full-text examination. Each research study was then categorized according to the topic covered (Table 1).

Table 1. Studies included in this review.

Topic	Subtopic	Authors (Year) [Reference]	Conclusions
Administrative applications		Eggmann, F., et al. (2023) [25]	ChatGPT can streamline administrative workflows and aid in clinical decision support, given comprehensive, unbiased data. However, it raises privacy and cybersecurity concerns and lacks reliability and up-to-date knowledge compared to traditional search engines, especially for health queries.
Clinical applications	<i>Diagnostics and radiology</i>	Baig, Z., et al. (2024) [26]	AI programs' treatment recommendations generally matched the current literature with up to 75% agreement, though data sources were often missing, except for Bard. Both GPT-4 and clinician reviews suggested procedures potentially leading to overtreatment. GPT-4 had the highest overall accuracy.
Clinical applications	<i>Diagnostics and radiology</i>	Islam, A., et al. (2024) [27]	The proficiency of ChatGPT in handling intricate reasoning queries within pathology demonstrated a noteworthy level of relational accuracy. Consequently, its text output created coherent links between elements, producing meaningful responses. This suggests that scholars or students can rely on this program to address reasoning-based inquiries.
Clinical applications	<i>Diagnostics and radiology</i>	Mohammad-Rahimi, H., et al. (2024) [28]	GPT-4 excelled in providing high-quality information on controversial dental topics. However, developers should incorporate scientific citation authenticators to validate citations due to the high incidence of fabricated references.
Clinical applications	<i>Diagnostics and radiology</i>	Russe, M.F.D.M., et al. (2024) [29]	A content-aware chatbot using GPT-4 reliably provided recommendations according to current consensus guidelines. The responses were deemed trustworthy and transparent and therefore facilitate the integration of artificial intelligence into clinical decision-making.
Clinical applications	<i>Diagnostics and radiology</i>	Sahu, P.K., et al. (2023) [30]	A content-aware chatbot using GPT-4 reliably followed current guidelines, providing trustworthy and transparent recommendations. This supports AI's integration into clinical decision-making.
Clinical applications	<i>Diagnostics and radiology</i>	Shikino, K., et al. (2024) [31]	ChatGPT-4 demonstrates potential as an auxiliary tool for diagnosing typical and mildly atypical presentations of common diseases. However, its performance declines with greater atypicality.
Clinical applications	<i>Diagnostics and radiology</i>	Silva, T.P., et al. (2024) [32]	The performance of the GPT program in describing and providing differential diagnoses for radiolucent lesions in panoramic radiographs is variable and at this stage limited in its use for clinical application.
Clinical applications	<i>Endodontics</i>	Mohammad-Rahimi, H., et al. (2024) [33]	GPT-3.5 provided more credible information on topics related to endodontics compared to Google Bard and Bing.

Table 1. Cont.

Topic	Subtopic	Authors (Year) [Reference]	Conclusions
Clinical applications	Endodontics	Ourang, S.A., et al. (2024) [34]	The paper reviews AI concepts in endodontics, focusing on machine learning for diagnosis and computer vision for dental image interpretation. It emphasizes the need for rigorous validation and ethical transparency. AI has significant potential to enhance endodontic research, education, and patient care with interdisciplinary collaboration.
Clinical applications	Endodontics	Qutieshat, A., et al. (2024) [35]	The study reveals AI's capability to outperform dental students in diagnostic accuracy regarding endodontic assessments.
Clinical applications	Endodontics	Snigdha, N.T., et al. (2024) [36]	The results showed no statistically significant differences between the two versions, indicating comparable response accuracy.
Clinical applications	Endodontics	Suarez, A., et al. (2024) [37]	The answers generated by ChatGPT showed high consistency (85.44%). ChatGPT achieved an average accuracy of 57.33%. However, significant differences in accuracy were observed based on question difficulty, with lower accuracy for easier questions.
Clinical applications	Oral and maxillofacial surgery	Acar, A.H. (2024) [38]	ChatGPT excels in answering oral surgery-related questions with superior accuracy, completeness, and clarity, making it a valuable tool for detailed information.
Clinical applications	Oral and maxillofacial surgery	Alten, A., et al. (2023) [39]	ChatGPT-4 can provide valuable information and guidance during orthognathic surgery consultations, but it cannot replace direct medical consultation.
Clinical applications	Oral and maxillofacial surgery	Balel, Y. (2024) [40]	Scholar GPT excelled in oral and maxillofacial surgery questions, providing more consistent and high-quality responses compared to ChatGPT. Models using academic databases offer more accurate and reliable information.
Clinical applications	Oral and maxillofacial surgery	Cai, Y., et al. (2024) [41]	ChatGPT/GPT-4 excelled in medical knowledge accuracy and recommendation rationality while also accurately sensing and providing reassurance about patient emotions. It can be used for patient follow-up after oral surgeries but should be supervised by healthcare professionals to consider current limitations.
Clinical applications	Oral and maxillofacial surgery	Çoban, E., and B. Altay (2024) [42]	The AI platform can educate patients about dental implantology and treatment procedures, but there is concern about potential bias toward specific dental implant brands.
Clinical applications	Oral and maxillofacial surgery	Isik, G., et al. [43]	The study outcomes emphasized high accuracy and quality in ChatGPT Plus's responses except for the questions requiring a detailed response or a comment.
Clinical applications	Oral and maxillofacial surgery	Jacobs, T., et al. (2024) [44]	AI was able to provide mostly accurate responses, and content was closely aligned with AAOMS guidelines. However, responses were too complex for the average third molar extraction patient and were deficient in citations and references.
Clinical applications	Oral and maxillofacial surgery	Suarez, A., et al. (2024) [37]	Final grade accuracy was found to be 71.7%, and consistency of the experts' grading across iterations ranged from moderate to almost perfect.
Clinical applications	Orthodontics	Abu Arqub, S., et al. (2024) [45]	The accuracy of ChatGPT's responses was generally insufficient, often missing relevant literature citations. Additionally, its capability to provide up-to-date and precise information was limited.
Clinical applications	Orthodontics	Daraqel, B. a. b., et al. (2024) [46]	Both ChatGPT- and Google Bard-generated responses were rated with a high level of accuracy and completeness to the general orthodontic questions posed. However, acquiring answers was generally faster using the Google Bard model.

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Topic	Subtopic	Authors (Year) [Reference]	Conclusions
Clinical applications	Orthodontics	Dursun, D., and R. Bilici Geçer (2024) [47]	All chatbot models provided generally accurate, moderately reliable, and moderate- to good-quality answers to questions the clear aligners.
Clinical applications	Orthodontics	Lima, N.G.M., et al. (2024) [48]	AI improves patient communication, diagnosis support, data digitization, and treatment assistance. ChatGPT aids in care, billing, and health information access but may provide nonsensical responses and poses privacy risks.
Clinical applications	Orthodontics	Makrygiannakis, M.A., et al. (2024) [49]	LLMs hold promise for evidence-based orthodontics, but their limitations can lead to incorrect decisions if not used carefully. They cannot replace orthodontists' critical thinking and expertise.
Clinical applications	Orthodontics	Surovková, J., et al. (2023) [50]	The paper introduces an AI-powered orthodontic workflow, highlighting new responsibilities for orthodontic nurses and assessing its use over three years with Dental Monitoring. It concludes that AI enhances dental practice with precise, personalized treatment but raises new ethical and legal issues.
Clinical applications	Patients' communication and self-education	Batool, I., et al. (2024) [51]	Embedded GPT model showed better results as compared to ChatGPT in providing postoperative dental care emphasizing the benefits of embedding and prompt engineering.
Clinical applications	Patients' communication and self-education	Gugnani, N., et al. (2024) [52]	Overall, the responses were found to be complete and logical and in clear language, with only some inadequacies being reported in a few of the answers.
Clinical applications	Patients' communication and self-education	Hassona, Y. (2024) [53]	ChatGPT is an attractive and potentially useful resource for informing patients about early detection of oral cancer. Nevertheless, concerns do exist about readability and actionability of the offered information.
Clinical applications	Patients' communication and self-education	Incerti Parenti, S., et al. (2024) [54]	The study suggests that while ChatGPT-3.5 can be a valuable tool for patient education, efforts to improve readability are necessary to ensure accessibility and utility for all patients.
Clinical applications	Patients' communication and self-education	Vassis, S., et al. (2024) [55]	Although patients generally prefer AI-generated information regarding the side effects of orthodontic treatment, the tested prompts fall short of providing thoroughly satisfactory and high-quality education to patients.
Clinical applications	Patients' communication and self-education	Yurdakurban, E., et al. (2024) [56]	AI-based chatbots with a variety of features have usually provided answers with high quality, reliability, and difficult readability to questions.
Clinical applications	Pediatrics	Rokhshad, R., et al. (2024) [57]	In the pilot study, chatbots showed lower accuracy than dentists. Chatbots may not be recommended yet for clinical pediatric dentistry.
Clinical applications	Periodontology	Alan, R., and B.M. Alan (2023) [58]	Consistently offered accurate guidance in most responses.
Clinical applications	Periodontology	Babayigit, O., et al. (2023) [59]	While ChatGPT may not offer absolute precision without expert supervision, it can still serve as a valuable resource for periodontologists, with some risk of inaccuracies.
Clinical applications	Periodontology	Danesh, A., et al. (2024) [60]	While ChatGPT 4 showed a higher proficiency compared to ChatGPT 3.5, both chatbot models leave considerable room for misinformation with their responses relating to periodontology.
Clinical applications	Periodontology	Tastan Eroglu, Z., et al. (2024) [61]	The present performance of ChatGPT in the classification of periodontitis exhibited a reasonable level. However, it is expected that additional improvements would increase its effectiveness and broaden its range of functionalities.

Table 1. Cont.

Topic	Subtopic	Authors (Year) [Reference]	Conclusions
Clinical applications	Prosthodontics	Freire, Y., et al. (2024) [62]	The results show that currently, ChatGPT has limited ability to generate answers related to RDPs and tooth-supported FDPs.
Clinical applications	Special needs	Rokhshad, R., et al. (2024) [63]	Chatbots exhibit acceptable consistency in responding to questions related to special needs dentistry and better accuracy in responding to true/false questions than diagnostic questions.
Clinical applications	Traumatology	Khan, M.K. (2024) [64]	AI and its subsets have been applied in a very limited number of fields of dental traumatology. However, the findings from the literature were found favorable and promising.
Clinical applications	Traumatology	Ozden, I., et al. (2024) [65]	Although ChatGPT and Google Bard are potential knowledge resources, their consistency and accuracy in responding to dental trauma queries remain limited.
Comprehensive		Alhaidry, H.M., et al. (2023) [66]	AI has greatly advanced dentistry, particularly in research. ChatGPT can transform dental and healthcare systems, but caution and policies are needed to mitigate hazards, and continuous monitoring is recommended due to ethical concerns and improper reference generation.
Comprehensive		de Souza, L.L., et al. (2024) [67]	Integrating ChatGPT in dentistry can be highly beneficial, but it is crucial to address ethical considerations, accuracy, and privacy concerns.
Comprehensive		Huang, H.Y., et al. (2023) [68]	While LLMs offer significant potential benefits, the challenges, such as data privacy, data quality, and model bias, need further study.
Dental research		Al-Moghrabi, D., et al. (2024) [69]	ChatGPT may effectively distinguish between predatory and legitimate journals, with accuracy rates of 92.5% and 71%, respectively.
Dental research		Bagde, H., et al. (2023) [70]	ChatGPT has the ability to provide appropriate solutions to questions in the medical and dentistry areas, but researchers and doctors should cautiously assess its responses because they might not always be dependable.
Dental research		Demir, G.B., et al. (2024) [71]	Both ChatGPT 3.5 and 4 can be pivotal tools for generating PICO-driven queries in orthodontics when optimally configured. However, the precision required in medical research necessitates a judicious and critical evaluation of LLM-generated outputs, advocating for a circumspect integration into scientific investigations.
Dental research		Fatani, B. (2023) [72]	ChatGPT can help find and summarize academic papers, generate drafts, and translate content, streamlining and simplifying academic writing. However, its use in scientific writing should be regulated and monitored due to ethical considerations.
Dental research		George Pallivathukal, R., et al. (2024) [73]	The study aids in creating guidelines for implementing GAI chatbots in healthcare education, emphasizing benefits and risks, and informing AI developers and educators about ChatGPT's potential in academia.
Dental research		Tiwari, A., et al. (2023) [74]	Studies show ChatGPT helps in scientific and dental research but should not be solely relied on due to ethical concerns and the need for review.
Dental research		Uribe, S.E., and I. Maldupa (2024) [75]	GenAI can potentially increase productivity and inclusivity, but it raises concerns such as bias, inaccuracy, and distortion of academic incentives. Therefore, the findings support the need for clear AI guidelines and standards for academic publishing.

Table 1. Cont.

Topic	Subtopic	Authors (Year) [Reference]	Conclusions
Educational enhancements		Claman, D., and E. Sezgin (2024) [76]	LLMs can enhance dental education by offering personalized feedback, case scenarios, and educational content. However, they also present challenges like bias, inaccuracies, privacy issues, and the risk of overreliance.
Educational enhancements		Roganović, J. (2024) [77]	A majority of students in the cohort were reluctant to use ChatGPT. Furthermore, familiarity (reading) with ChatGPT features appears to alter the expectations and enhance learning performance of students, suggesting an AI description-related cognitive bias.
Educational enhancements	General expertise	Albagieh, H., et al. (2024) [78]	No significant difference was found in response scores. However, residents showed low agreement, while LLMs showed high agreement. Dentists should leverage AI for diagnosis and treatment.
Educational enhancements	General expertise	Ali, K., et al. (2024) [79]	Generative AI can transform virtual learning. Healthcare educators should adapt to its benefits for learners while mitigating dishonest use.
Educational enhancements	General expertise	Aminoshariae, A., et al. (2024) [80]	AI in endodontic education will support clinical and didactic teaching through individualized feedback; enhanced, augmented, and virtually generated training aids; automated detection and diagnosis; treatment planning and decision support; and AI-based student progress evaluation and personalized education.
Educational enhancements	General expertise	Giannakopoulos, K., et al. (2023) [81]	Although LLMs show promise in evidence-based dentistry, their limitations can lead to harmful decisions if not used carefully. They should complement, not replace, a dentist's critical thinking and expertise.
Educational enhancements	General expertise	Kunzle, P., and S. Paris (2024) [82]	Overall, there are large performance differences among LLMs. Only the ChatGPT-4 models achieved a success ratio that could be used with caution to support the dental academic curriculum.
Educational enhancements	General expertise	Li, C., et al. (2024) [83]	For periodontal surgery exams, ChatGPT's accuracy was not as high as students', but it shows potential in assisting with the curriculum and helping with clinical letters and reviews.
Educational enhancements	General expertise	Molena, K.F., et al. (2024) [84]	ChatGPT initially demonstrated good accuracy and completeness, which was further improved with machine learning (ML) over time. However, some inaccurate answers and references persisted.
Educational enhancements	General expertise	Praveen, G., et al. (2024) [85]	ChatGPT generated clear, scientifically accurate and relevant, comprehensive, and consistent responses to diverse oral health-related queries despite some significant limitations.
Educational enhancements	General expertise	Puladi, B., et al. (2024) [86]	Classic OMS diseases are underrepresented. The current literature related to LLMs in OMS has a limited evidence level.
Educational enhancements	General expertise	Sabri, H., et al. (2024) [87]	ChatGPT-4 performed well on AAP in-service exam questions, outperforming Gemini and ChatGPT-3.5. While it shows potential as an educational tool in periodontics and oral implantology, limitations like processing image-based inquiries, inconsistent responses, and not reaching absolute accuracy must be considered.
Educational enhancements	GPT vs. literature research	Kavadella, A., et al. (2024) [88]	Students using ChatGPT for their learning assignments performed significantly better in the knowledge examination than their fellow students who used the literature research methodology.
Educational enhancements	GPT vs. literature research	Saravia-Rojas, M.A., et al. (2024) [89]	Dental students highly valued the experience of using ChatGPT for academic tasks. Nonetheless, the traditional method of searching for scientific articles yield higher scores.

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Educational enhancements	Licensing exam/mastery tests	Chau, R.C.W., et al. (2024) [90]	The newer version of GenAI has shown good proficiency in answering multiple-choice questions from dental licensing examinations.
Educational enhancements	Licensing exam/mastery tests	Dashti, M., et al. (2024) [91]	Both ChatGPT 3.5 and 4 effectively handled knowledge-based, case history, and comprehension questions, with ChatGPT 4 being more reliable and surpassing the performance of 3.5. ChatGPT 4's perfect score in comprehension questions underscores its trainability in specific subjects. However, both versions exhibited weaker performance in mathematical analysis.
Educational enhancements	Licensing exam/mastery tests	Farajollahi, M., and A. Modaberi (2023) [92]	Out of 100 questions asked from ChatGPT, a score of 40 was obtained.
Educational enhancements	Licensing exam/mastery tests	Fuchs, A., et al. (2023) [93]	The performance disparity between SFLEDM and EEAACI assessments highlights ChatGPT's varying proficiency due to differences in training data. Priming can help, but healthcare use must be cautious due to inherent risks.
Educational enhancements	Licensing exam/mastery tests	Jeong, H., et al. (2024) [94]	The performance of chatbots in oral and maxillofacial radiology was unsatisfactory. Further training using specific, relevant data derived solely from reliable sources is required.
Educational enhancements	Licensing exam/mastery tests	Jin, H.K., et al. (2024) [95]	The accuracy levels ranged from 36 to 77% for ChatGPT-3.5 and from 64.4 to 100% for GPT-4. Additionally, in the context of health licensing examinations, the ChatGPT models exhibited greater proficiency in the following order: pharmacy, medicine, dentistry, and nursing.
Educational enhancements	Licensing exam/mastery tests	Kim, W., et al. (2024) [96]	Using the KDLE as a benchmark, the study demonstrates that although LLMs have not yet reached human-level performance in overall scores, both Claude3-Opus and ChatGPT-4 exceed the cut-off scores and perform exceptionally well in specific subjects.
Educational enhancements	Licensing exam/mastery tests	Morishita, M., et al. (2024) [97]	While innovative, ChatGPT-4V's image recognition feature exhibited limitations, especially in handling image-intensive and complex clinical practical questions, and is not yet fully suitable as an educational support tool for dental students at its current stage.
Educational enhancements	Licensing exam/mastery tests	Ohta, K., and Ohta, S. (2023) [98]	GPT-4 achieved the highest overall score in the JNDE, followed by Bard and GPT-3.5. However, only Bard surpassed the passing score for essential questions.
Educational enhancements	Licensing exam/mastery tests	Revilla-León, M., et al. (2024) [99]	The AI-based chatbot tested not only passed the exam but performed better than licensed dentists.
Educational enhancements	Licensing exam/mastery tests	Song, E.S., and S.P. Lee (2024) [100]	GPT-4 shows great potential for medical education and standardized testing, especially in English. However, performance varies across subjects and languages, highlighting the need for diverse and localized training data to improve effectiveness.
Educational enhancements	Licensing exam/mastery tests	Takagi, S., et al. (2024) [101]	ChatGPT-4V successfully passed the 117th JMLE, demonstrating proficiency in handling questions, including image- and table-based questions.
Educational enhancements	The students' perspective	Abdaljaleel, M., et al. (2024) [102]	The study validated "TAME-ChatGPT" as a useful tool for assessing ChatGPT adoption among university students.
Educational enhancements	The students' perspective	Alnaim, N., et al. (2024) [103]	Despite challenges and varied perceptions based on gender and education level, the overwhelmingly positive attitudes toward ChatGPT underscore its potential as a valuable tool in medical education.

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Topic	Subtopic	Authors (Year) [Reference]	Conclusions
Educational enhancements	<i>The students' perspective</i>	Kurt Demirsoy, K., et al. (2024) [104]	ChatGPT has significant potential in terms of usability for patient information and education in the field of orthodontics if it is developed and necessary updates are made.
Educational enhancements	<i>The students' perspective</i>	Rahad, K., et al. (2024) [105]	The results showed that ChatGPT can assist in dental essay writing and generate relevant content for dental students, in addition to other benefits.
Educational enhancements	<i>The students' perspective</i>	Sallam, M., et al. (2023) [106]	ChatGPT has potential in medical, dental, pharmacy, and public health education by improving personalized learning, clinical reasoning, and the understanding of complex concepts. However, concerns include data privacy, biased and inaccurate content, and risks to critical thinking and communication skills, highlighting the need for proper guidelines.
Educational enhancements	<i>The teachers' perspective</i>	Ahmed, W.M., et al. (2024) [107]	ChatGPT and Bard can generate numerous questions about dental caries, especially at the knowledge and comprehension levels, making them useful for large-scale exams. However, educators need to review and adapt these questions to ensure they meet their learning objectives.
Educational enhancements	<i>The teachers' perspective</i>	Brondani, M., et al. (2024) [108]	Instructors could usually tell if reflections were generated by ChatGPT or students. However, the thematic analysis content from ChatGPT matched that of qualitative researchers.
Educational enhancements	<i>The teachers' perspective</i>	de Vries, T.J., et al. (2024) [109]	These methods proved to be appropriate and logical choices for reaching the learning goals of the course.
Educational enhancements	<i>The teachers' perspective</i>	Quah, B., et al. (2024) [110]	The study shows the potential of ChatGPT for essay marking. However, an appropriate rubric design is essential for optimal reliability.
Educational enhancements	<i>The teachers' perspective</i>	Shamim, M.S., et al. (2024) [111]	AI and ML technologies can potentially supplement human grading in the assessment of essays.
Educational enhancements	<i>The teachers' perspective</i>	Shete, A., et al. (2024) [112]	Instead of treating artificial intelligence as a threat, dental educators need to adapt teaching and assessments in dental education for the benefit of learners while mitigating its dishonest use.
Educational enhancements	<i>The teachers' perspective</i>	Uribe, S.E., et al. (2024) [113]	A positive yet cautious view towards AI chatbot integration in dental curricula is prevalent, underscoring the need for clear implementation guidelines.

4. Discussion

Currently, ChatGPT is available in a free version as ChatGPT 3.5 and a more advanced version that requires a subscription as ChatGPT 4. Technological differences between the two versions can affect the output from the chatbox. Hirosawa T. et al. assessed the accuracy of differential diagnoses generated by ChatGPT 3.5 and ChatGPT 4, comparing the results with those from human doctors to evaluate their alignment with the correct final diagnosis. The analysis revealed that ChatGPT 4 achieved higher rates of correct diagnosis compared to ChatGPT 3.5 [114].

Using one version or the other for any type of application could therefore yield different results. The advantages of version 4 are listed below:

- It can process a greater number of words simultaneously compared to ChatGPT 3.5, enabling more extensive conversations and context as well as improved text comprehension.
- It provides more accurate responses thanks to a larger and more diverse database.
- It successfully incorporates contextual details and generates responses that are consistent not just with the immediate input but with an entire conversation.

- It reduces the number of plausible but incorrect responses (known as “hallucinations”), providing a crucial improvement in the trustworthiness of the generated results.

Although version 4 manages to reduce the phenomenon of hallucinations, this issue is still significantly present, potentially compromising the reliability of the responses [115]. One of the major inherent limitations of this technology is indeed its tendency to produce outputs that are not based on truth or objective reality but rather plausible responses. ChatGPT 4 defines artificial hallucination as “the production of information that seems plausible but is incorrect or fabricated. It occurs when the model generates a response based on patterns it has learned during training, even though the information may not be factually accurate or may not exist in real-world knowledge. This can be particularly challenging in situations where accuracy is critical, such as medical advice or factual reporting. Efforts are continuously made to reduce the frequency of these hallucinations by improving the training data and the models’ algorithms.” Hence, the human element remains crucial in analyzing the chatbot’s response and in evaluating its truthfulness, as it cannot be given complete autonomy.

Despite this inherent limitation in GPT technology, the chatbot has proven useful in all fields of medicine as a support tool. This narrative review assesses the areas where ChatGPT can aid the dental sector, regardless of the version used. The literature review has highlighted various areas of dentistry where ChatGPT proves to be particularly effective.

4.1. Applications in Dental Research

Uribe and Maldupa systematically assessed the use of generative AI in dental research, showing how from 2018 to 2024, the use of words signaling the employment of GPTs in dental research abstracts increased from 47.1 per 10,000 papers to 224.2 per 10,000 papers [75]. Due to the language focus of LLM technology, it has significant potential for application in writing, reviewing, and editing research.

Several authors showed how this tool is effective in increasing productivity, inclusivity, and overall help and at the same time raised strong research ethics concerns, often promoting the need of strict guidelines [70,73,74]. Nonetheless, ChatGPT streamlines the synthesis of the extensive dental literature, facilitating systematic reviews by suggesting PICO questions and meta-analyses by identifying gaps and suggesting future research directions, thus enhancing knowledge discovery and evidence-based practice, or generating figures, tables, and other visual elements to summarize data [71,72]. ChatGPT currently does not have direct access to databases of scientific articles such as PubMed, Embase, Web of Science, or Google Scholar; the articles to be included in the review need to be manually added into the chatbox. Hence, the inclusion phase in the review remains a task that requires the operator’s choice and not that of the AI. On the other hand, the subsequent phases of screening and eligibility can be performed by AI.

The use of ChatGPT in research proves useful in correcting sentences with syntactic or grammatical errors and translating them into other languages or in setting up the bibliography according to a well-defined format. However, it cannot be used to generate bibliographic references. Numerous studies suggest that references generated by ChatGPT may indeed be fictitious [4,116–118]. In the submission phase, ChatGPT seems to effectively distinguish between predatory and legitimate dental journals, with accuracy rates of 92.5% and 71%, respectively [69].

4.2. Clinical Applications

4.2.1. Diagnostics and Radiology

Although it is not possible to delegate complete diagnostic autonomy, ChatGPT can assist clinicians by providing information on possible diagnoses if symptom descriptions are inputted. ChatGPT-4 demonstrates potential as an auxiliary tool for diagnosing typical and mildly atypical presentations of common diseases, can reliably provide trustworthy and transparent recommendations according to current consensus guidelines, and can clarify doubts [27–30]. When looking for treatment recommendations, LLMs generally

aligned with the current literature up to 75% of the time, although data sources were frequently missing. Both GPT-4 and clinician reviews proposed procedures that might lead to overtreatment. Among the models, GPT-4 demonstrated the highest overall accuracy [26]. However, its performance has been shown to decline with greater atypicality [31].

The performance of the GPT program in describing and providing differential diagnoses for radiolucent lesions in panoramic radiographs is variable and at this stage limited in its use for clinical application [32]. When inputting into the ChatGPT 4 chatbox “If I provide you with images of intraoral X-rays, or orthopantomograms, would you be able to diagnose pathologies such as tooth decay, periodontitis, or other oral pathologies?” the chatbot responds: “I’m not able to diagnose medical conditions or interpret medical images like intraoral X-rays. It’s important to consult a qualified healthcare professional, such as a dentist or radiologist, who can provide accurate diagnoses and appropriate treatment recommendations for tooth decay, periodontitis, or other oral pathologies.” It is not possible to use the ChatGPT 4 chatbox directly as a diagnostic tool. Within the OpenAI GPT platform, it is possible to find numerous pre-trained AIs, or to train a new one, to overcome this limitation.

4.2.2. Traumatology

Regarding dental traumatology, AI and LLMs have been limitedly applied. Ozden et al. stated in their research that although ChatGPT and Google Bard are potential knowledge resources, their consistency and accuracy in responding to dental trauma queries remain limited, with correct answers to 57.5% of the questions [64,65].

4.2.3. Oral and Maxillofacial Surgery

Employment of ChatGPT as an intelligent virtual assistant in oral surgery has been tested, with accuracy results of 71.7%, and response consistency from moderate to almost perfect and good proficiency as a consultant on the subject [38,119]. It was shown to provide valuable information and guidance during orthognathic surgery consultations [39]. Moreover, it was able to assist in patient communication for third molar extraction, dental implantology, and after-surgery follow-ups [41,42,44]. However, Çoban, E., and B. Altay expressed their concerns about potential bias toward specific dental implant brands in these answers [42].

While Isisk et al. described high accuracy and quality in ChatGPT Plus’s responses in the field of oral and maxillofacial surgery, except for the questions requiring a detailed response or a comment, Balel et al. used a modified version named ScholarGPT. The results excelled in oral and maxillofacial surgery questions, providing more consistent and high-quality responses compared to ChatGPT [40,43]. These results suggest that GPT models based on academic databases can provide more accurate and reliable information. Scholar GPT can in fact access a wide range of scholarly resources, including Google Scholar, PubMed, JSTOR, and Arxiv, and it has built-in critical reading skills to help analyze and interpret research papers and PDFs.

4.2.4. Prosthodontics

Freire et al. evaluated the performance of ChatGPT in generating answers about removable dental prostheses (RDPs) and tooth-supported fixed dental prostheses (FDPs). Technical questions were formulated regarding the various stages of prosthesis fabrication, including impressions, vertical dimension, trial phases, occlusal registration, and cementation. The answers showed a reliability of 25.6%, demonstrating that currently, ChatGPT has a limited ability to generate answers related to RDPs and tooth-supported FDPs [62]. The prosthodontist cannot rely on the chatbot to obtain information on correctly making a prosthesis yet.

4.2.5. Periodontology

Regarding periodontal diseases, LLMs seem to consistently offer accurate guidance in most responses and can classify periodontitis at a reasonable level [58,59,61].

Danesh et al., on the other hand, reported that ChatGPT 3.5 and ChatGPT 4 answered 57.9% and 73.6% of in-service questions correctly. They concluded that while ChatGPT 4 showed a higher proficiency compared to ChatGPT 3.5, both chatbot models leave considerable room for misinformation with their responses relating to periodontology [60].

4.2.6. Endodontics

On the quality of endodontic information provided, ChatGPT outperforms dental students in diagnostic accuracy regarding endodontic assessments, and its 3.5 version seems to provide more credible information on topics related to endodontics compared to Google Bard and Bing, although it only gives 60% correct answers [33–36]. Suarez et al. simulated patient–physician questions related to the field of endodontics. The answers generated by ChatGPT showed high consistency (85.44%) and an average accuracy of 57.33%. However, significant differences in accuracy were observed based on question difficulty, with lower accuracy for easier questions [37].

4.2.7. Orthodontics and Pediatrics

LLMs seem to show great potential in supporting evidence-based orthodontic choices, data digitization, and treatment assistance with a high level of accuracy and completeness to the general orthodontic questions [46,48,49]. Dursun and Bilici Geçer in their work explained how generally accurate, moderately reliable, and moderate- to good-quality answers to questions about clear aligners were given [47]. On the other hand, Abu Arqub et al. demonstrated that ChatGPT has a limited ability to provide completely accurate and reliable responses regarding aligners: 58% of the AI-generated answers were scored as objectively true, 18% were selected facts, 9% were minimal facts, and 15% were false [45].

AI-powered orthodontic workflow was described by Surokova et al., who concluded that it enhances dental practice with precise, personalized treatment and at the same time raises new ethical and legal issues [50].

Rokhshad et al., in two separate studies on special needs and pediatric patients, demonstrated that chatbots showed acceptable consistency for the former group but lower accuracy than dentists for the latter [57,63].

4.2.8. Patients' Communication and Self-Education

As this technology becomes widely and freely available, it is to be expected that patients will try to inform themselves using these tools, as they began to do when internet search engines became widespread.

Incerti Parenti et al. in his research regarding obstructive sleep apnea showed how ChatGPT-3.5 delivers higher quality and more comprehensive information compared to Google Search, although its responses are less readable [54].

This AI model excels in customizing patient communications, delivering understandable explanations of dental conditions and treatments, postprocedural instructions, and responses to frequently asked questions, thereby enhancing patient engagement and compliance. This can improve patient understanding and compliance with treatment. This possible application has been confirmed in the fields of orthognathic surgery, orthodontics, oral cancer, pediatrics, and postoperative care with overall good results [51–53,55,56]. Vassis et al. rated as “neither deficient nor satisfactory” the information received from ChatGPT, with GPT-4 achieving higher scores than GPT-3.5. At the same time, the patients perceived the GPT-generated information as more useful and more comprehensive and experienced less nervousness when reading the GPT-generated information, preferring in 80% of cases the AI-generated information over the standard text [55].

4.3. Administrative Applications

ChatGPT can help create precise and exhaustive patients' documentation. By entering patient data, clinical findings, and treatment plans, specialists can utilize ChatGPT's capabilities for text generation to formulate comprehensive reports [68]. This method boosts the quality of documentation, ensuring that critical information is effectively recorded and may support seamless communication among healthcare providers, possibly overcoming language barriers in doctor–patient interaction. While no specific result regarding administrative application in dentistry resulted from the research selection, Eggmann et al. underlined the potential of LLMs in supporting health care professionals in routine written communications and record keeping, potentially enhancing the quality of patient care and saving time, thus reducing costs [25]. However, the input of sensitive health-related information raises privacy and cybersecurity concerns.

4.4. Educational Enhancements

ChatGPT can be effectively employed to improve the dentistry educational field for both students and teachers.

4.4.1. The Students' Perspective

Abdaljaleel et al. conducted the TAME-ChatGPT (Technology Acceptance Model Edited to Assess ChatGPT Adoption) survey among university students for educational purposes [102]. The results showed that the employment of this LLM is supported due to increased accuracy and speed, ease of use, perceived usefulness, low perceived risks, and low "anxiety" levels among the participants, suggesting a readiness to adopt ChatGPT despite the recognized concerns. The study found that students generally had low "anxiety" about potential drawbacks of ChatGPT, such as declining critical thinking, overdependence on technology, and diminished originality. Instead, they viewed ChatGPT as a valuable educational tool with low perceived "anxiety" about academic integrity or skill development issues. Students' positive attitude towards the use of LLMs and usefulness is also confirmed by several other authors [103–106].

4.4.2. The Teachers' Perspective

A worldwide survey of educators by Uribe et al. showed a prevalent positive yet cautious view towards AI chatbot integration in dental curricula, underscoring the need for clear implementation guidelines [113]. GPTs can generate always-changing questions for exams, especially at the knowledge and comprehension levels, making them useful for large-scale exams provided that supervision on the output is always ensured [107].

Moreover, ChatGPT may generate scenarios, questions, answers, and explanations in real time, building more engaging learning for students. This type of content can simulate a realistic conversation between students and virtual patients or clinical scenarios, offering the chance to exercise in a controlled environment. de Vries et al. included ChatGPT actively in an entire course, deeming it extremely useful also in summarizing general overarching principles, which are sometimes spread over more than one chapter, allowing students to reach the learning goals [109].

Educators can use GPTs to check if tasks were fulfilled using GPTs, adjusting their teaching and assessments to benefit learners while addressing potential misuse [108,112]. Moreover, GPTs can supplement humans in grading paper and essays [110,111].

4.4.3. Mastery and Expertise Tests

Several authors tested LLMs' capabilities against licensing exams and specialist boards. Their good proficiency was underlined in multiple-choice questions and short answers rather than thematic, open ones or mathematical analysis. The results exceeded the cut-off scores and showed they performed exceptionally well in specific subjects, in one case performing better than licensed dentists, as described by Revilla-Léon et al., with the exception of the Iranian Endodontics Specialist Board, where the tool employed obtained a

score of 40/100 [90–93,95–101]. Another underwhelming result was achieved in oral and maxillofacial surgery. Specifically, in Jeong et al.'s study, the students' overall accuracy rate was 81.2%, while that of the chatbots varied from 50.0% to 65.4%. ChatGPT Plus achieved a higher accuracy rate for basic knowledge than the students (93.8% vs. 78.7%). However, all chatbots performed poorly in image interpretation, with accuracy rates below 35.0% [94]. On the topic of image recognition, Morishita et al. showed how the new ChatGPT-4V's image recognition feature exhibited limitations, especially in handling image-intensive and complex clinical practical questions, and is not yet fully suitable as an educational support tool for dental students at its current stage [97].

Tests were also performed to assess the proficiency against patients' oral health-related queries, in-service exam questions, oral lesion diagnosis, and recognized assessments in several dental subjects with good outcomes [78–85,87]. Regarding oral and maxillofacial surgery, Puladi et al. highlighted that classic diseases are underrepresented, and the current literature on large language models in this field lacks sufficient evidence [86].

In the dispute between GPT use vs. literature research methodology, Kavadella et al. showed how students using ChatGPT to compose a learning assignment performed significantly better than their fellow students who used the literature research methodology, while in Saravia-Rojas et al.'s work, higher values through the traditional method than with ChatGPT were reached regarding final scores and scores for the criteria of utilization of evidence, evaluation of arguments, and generation of alternatives. Nonetheless, the students highly valued the experience of using ChatGPT for academic tasks [88,89].

Shortcomings of this technology in the educational field have been stressed by authors regarding cognitive and general bias, inaccuracies, privacy issues, and the risk of overreliance [76,77].

4.5. Ethical and Practical Considerations

Legal issues related to the use of ChatGPT are troubling due to unclear accountability and copyright matters [116,120,121]. Fighting plagiarism and academic incorrectness requires the creation of solid ethical principles that harmonize the usefulness of AI with personal interests. Researchers who use ChatGPT for writing academic papers should openly disclose its use.

There are numerous problems related to patient privacy, which could be compromised [67,121]. When sensitive patient information is entered into the ChatGPT chatbox, it is unclear how these data are stored or for how long. Even if it is unintentional, data may be stored on the AI's servers, increasing the risk of unauthorized access. These data could be used to further train the AI model. This means that sensitive information could be used in ways that were not intended or authorized, including sharing with third parties. Any computer system, including those hosting ChatGPT, can be vulnerable to security breaches. In the event of a breach, sensitive patient information could be exposed to bad actors. To protect patients' data, the use of offline LLMs such as META LLaMA or any kind of anonymization could be a solution. Nonetheless, offline LLMs should be proposed as standard, since anonymization still raises possible concerns due to the data being sent and processed to external servers anyway [68]. This use also lacks compliance with privacy regulations: using AI to manage patient data may not comply with privacy regulations such as the GDPR in Europe or HIPAA in the United States, if the AI is not properly configured to protect and manage data in compliance with these laws. To mitigate these risks, it is essential that any use of AI, such as ChatGPT, in a clinical context is carefully evaluated and managed [122,123].

At present (November 2024), ChatGPT is accessible at no cost in its GPT-4o mini version and for the use of custom GPTs. Monthly fees are charged for updated and multimodal builds (also known as GPT-4, GPT-4o, and DALL-E) and to create customized GPT versions. Yet, there exists a concern that ChatGPT and comparable artificial intelligence systems could become prohibitively costly to access as their popularity increases. This might lead to a disproportionate allocation of resources among researchers across

different disciplines [66,124]. Various publications have even credited the chatbot with authorship [125,126]. Nevertheless, attributing authorship carries significant ethical implications for scholarly articles and has emerged as a pressing issue in academic journals. A recent article in *Nature* declared that an AI chatbot should not be recognized as author of a scientific paper, as it cannot be accountable for the claims made in the article [127].

4.6. Future Directions

As AI technologies continue to advance, ChatGPT is expected to become more integrated into dental practice, offering advanced diagnostic tools, personalized treatment options, and real-time predictive analytics.

It could become an even more effective tool to assist dentists in preliminary diagnoses and in providing personalized advice to patients; it could be trained with dentistry-specific datasets to recognize symptoms, suggest diagnostic tests, and offer preliminary treatment options based on the most up-to-date practices. As GPTs become more advanced, if a version specifically trained for dental offices were integrated into the management software, it could help with patient management, appointment scheduling, and even managing prescriptions and communications with other healthcare professionals. Despite notable successes, AI technology and specifically models like ChatGPT have numerous limitations. The quality and reliability of responses may be inconsistent, particularly on topics with a wide variety of opinions or less documented knowledge. Furthermore, these models can generate plausible but incorrect responses, a phenomenon known as “hallucination” in the generated texts [115].

These limitations raise significant concerns, especially in critical fields such as medicine, where the accuracy of information is crucial. To overcome these obstacles, AI developers are working on multiple fronts: improving learning algorithms to reduce the risk of errors, increasing the size and quality of training datasets to better cover the breadth and depth of human knowledge, and developing new methodologies for validating the responses generated to ensure their reliability. Furthermore, the implementation of transparency systems in AI answers is another crucial field of development, which will help build confidence in its use in professional systems and contexts, including dentistry and medicine in general.

4.7. Alternative AI Models to ChatGPT

There are several alternative language models to ChatGPT, each with unique characteristics: Anthropic’s Claude, Google’s Gemini (formerly known as Bard), Microsoft’s Copilot, Meta’s Llama, and BERT (and its derivatives such as ALBERT, RoBERTa, DistilBERT, SpanBERT, and DeBERTa) are just few of the available language models that can be used depending on the task to be performed. While GPT-4 is a generative model based on a unidirectional (auto-regressive) transformer architecture, BERT, RoBERTa, and ALBERT use a bidirectional architecture. This difference allows bidirectional models to better understand the context of words by simultaneously considering the words to the left and right of a given word. GPT-4, on the other hand, generates the text one word at a time sequentially, but it is optimized to produce coherent and fluent texts. Some models like Copilot are based on GPT-4 and enhanced with data coming from proprietary search engines like Bing or Google search. Generally speaking, every model is optimized for a variety of natural language processing tasks, each with unique approaches to transformer architectures and training techniques. Therefore, the choice of the best model depends on the usage context and specific application needs. Studies comparing various versions of the same model or different language models are increasingly appearing in scholarly publications, highlighting the growing interest in these models in dentistry and in other specific fields of medicine [49,71,98,128,129].

4.8. Limitations

At the time of conducting this review, there is limited original research on the application of ChatGPT in dentistry.

The most significant limitation in comparing the selected research results lies in the methods used. Different LLMs, or different versions of the same LLM, vary in the quality of their responses. This issue, along with the automatic “contextual refinement” performed by LLMs, makes comparisons between papers difficult, if not impossible, unless each material and methods section specifies the exact version and build of the software used and details any contextual refinement or prompt engineering applied, if any. Another limitation is the model’s tendency to generate “hallucinations”, that is, responses that seem plausible but are actually incorrect or invented. This phenomenon represents a significant challenge and can lead to erroneous diagnoses or inaccurate medical advice if the model is used without human supervision. To address this issue, various methods can be employed. One is fine-tuning on curated data, using high-quality datasets to ensure reliable information. Retrieval-augmented generation (RAG) combines the model’s generative abilities with external sources for more accurate responses. Model calibration adjusts prediction confidence levels for reliability. Prompt engineering guides models to better outputs by crafting precise questions and instructions. Human-in-the-loop feedback involves humans correcting hallucinations, aiding the model’s learning. Post-processing techniques filter out hallucinated content for more accurate final responses.

ChatGPT is not designed to interpret medical images such as intraoral X-rays or orthopantomographs, which limits its usefulness as a direct diagnostic tool. Another critical aspect is the necessity for constant system updates to keep the knowledge up-to-date with recent developments in the medical field, thereby avoiding the dissemination of outdated or incorrect information. The lack of direct access to scientific databases limits its ability to provide responses based on the updated medical literature, making the experts’ intervention indispensable to ensure the accuracy of the information used and generated by the model.

Prompt engineering, while a powerful tool for guiding AI systems, does come with its own set of challenges that might be seen as limitations to AI’s use and to study comparison. The end-user should be aware of the training datasets used, since their quality may impact the performance of the model or may present biases. The model may not behave as intended due to prompt formulation, or communication barriers may rise when it does not fully understand the intent or the subtle distinctions behind the prompts. It can be challenging to obtain the exact output you want from the AI model in the first attempt.

“Contextual understanding” or “contextual refinement” is when LLMs refine their answers using the context from earlier interactions to provide more accurate and relevant responses. If research studies do not account for this automatic adjustment, the results may be biased. Therefore, determining the best way to initiate a prompt, and in the end, understanding and evaluating the reasoning behind the generated responses can be tricky. These challenges highlight the complexity of prompt engineering and the need for careful consideration and ongoing refinement to achieve optimal, comparable results.

5. Conclusions

The increasing number of studies published in the literature over the past two years on ChatGPT, and large language models in general, show a real interest in this tool and its application in diverse areas of dentistry:

- Dental research: It has been shown to be useful in study designing, abstract generation, draft correction, syntax error correction, translations, and reference formatting. At present, ChatGPT cannot be trusted to generate bibliographic references from a text due to frequent “hallucinations”, errors, and/or outdated information.
- Clinical application: ChatGPT can be used, only if properly trained by the operator, to assist the diagnostic workflow. However, it is not yet possible to use it directly as a diagnostic tool, and answers should not be accepted without question, primarily because there is a chance that an outdated version of the model might be in use.

- **Administrative applications:** Through inputted clinical findings and treatment plans, the tool is able to generate high-quality reports, spot trends, and manage patient follow-up appointments.
- **Educational enhancements:** By simulating a realistic conversation, it is capable of assisting students in generating scenarios, questions, answers, and explanations in a quick and engaging way. It can compose ever-changing tests and grade them.
- **Ethical and practical considerations:** The processing and storage of patients' sensitive data raises significant privacy concerns. This is a crucial aspect to consider in the deployment of such tools in a clinical setting.

The results of this narrative review show embryonic results with little clinical application with patients; thus, ChatGPT is still far from being used on a daily basis. Nonetheless, ChatGPT stands as a pivotal innovation in dentistry, promising to enhance research capabilities, clinical accuracy, administrative efficiency, and educational processes. Effective integration, coupled with stringent ethical practices and ongoing validation, is essential for realizing the full potential of AI in dentistry.

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